A Predictive Model of Shot Peening Coverage

Swept-jet peening: influence of the number of passes, the media, the jet size and the traverse speed

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Abstract

This note explains the coverage model implemented in the spreadsheet `tauxderecouvrement.xlsx`. The model predicts the coverage τ of a surface treated by a swept jet of peening media as a function of the number of passes n, the media characteristics (flow rate, density, diameter), the jet size ΔX and the traverse speed V, for a given dent radius r. It relies on two steps: (i) a mass balance giving the number of impacts received per unit area at each pass, and (ii) a random-coverage law (Poisson law, also known as the Avrami law) converting this number of impacts into the fraction of surface covered.

1 Main result

The coverage after n passes is

$$\boxed{\tau(n) = 100\left(1 - e^{-K\,n}\right) \quad [\%]} \tag{1}$$

where K is the *fraction of surface "impacted" per pass* (overlaps not taken into account):

$$K = \pi r^2\, R \qquad \text{with} \qquad R = \frac{\dot{N}}{V\,\Delta X} \qquad \text{and} \qquad \dot{N} = \frac{\dot{m}}{\rho\,\frac{\pi}{6}D^3}. \tag{2}$$

Combining these gives the compact form:

$$\boxed{K = \frac{6\,\dot{m}\,r^2}{\rho\,D^3\,V\,\Delta X}} \tag{3}$$

With the values of the spreadsheet, $K \approx 0.337$: each pass deposits a cumulative dent area equal to about one third of the swept area.

2 Input parameters

Table 1: Model parameters and values in the spreadsheet.

Quantity	Symbol	Value	Unit	Excel cell
Media mass flow rate	$\dot{m}$	1.5	kg/min	B2 (B3 in kg/s)
Media density	ρ	2.2	g/cm^3	B4 (B5 in kg/m^3)
Shot diameter	D	150	µm	B6 (B7 in m)
Jet size	ΔX	20	mm	B11
Jet traverse speed	V	300	mm/s	B12
Dent radius	r	10	µm	B16
Number of passes	n	0 to 39	–	F3:F42

The dent radius r is an *input*: it is not computed by the model (in reality it depends on the impact velocity, the media and the hardness of the part, and is measured or estimated separately). This is why the model predicts coverage "for a given dent size".

3 Building the model step by step

3.1 Mass of one shot

Shots are assumed spherical with diameter D:

$$m_p = \rho \, \frac{4}{3}\pi \left(\frac{D}{2}\right)^3 = \rho \, \frac{\pi D^3}{6}. \tag{4}$$

Numerical value: $m_p = 2200 \times \frac{\pi}{6} \, (150 \times 10^{-6})^3 \approx 3.89 \times 10^{-9}$ kg, i.e. 3.89×10^{-3} mg (cells B8, B9).

3.2 Shot flow rate

The number of shots thrown per second is the mass flow rate divided by the mass of one shot:

$$\dot{N} = \frac{\dot{m}}{m_p}. \tag{5}$$

Numerical value: $\dot{N} = \dfrac{0.025}{3.89 \times 10^{-9}} \approx 6.43 \times 10^6$ shots/s (cell B10).

3.3 Impact density per pass

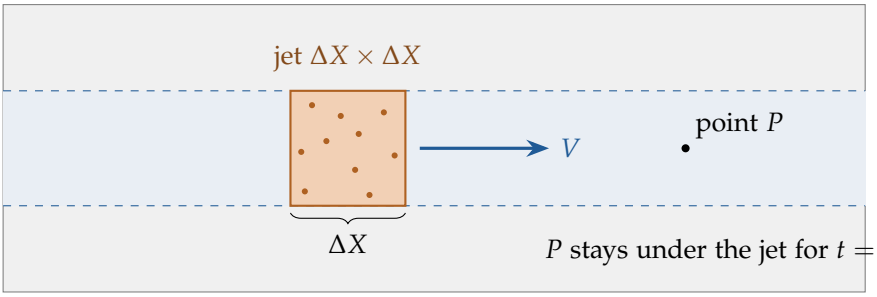


Figure 1: Sweep geometry: a square jet of side ΔX moves at speed V; one "pass" is one passage of the jet over a point of the part.

The jet is assumed to be a *square* spot of side ΔX (area ΔX^2) in which shots arrive uniformly. The impact flux under the jet is therefore $\dot{N}/\Delta X^2$ (impacts per second and per mm^2).

A point P of the part stays under the jet for the dwell time $t = \Delta X / V$ (figure 1). The mean number of impacts received per unit area during one pass is therefore:

$$R = \frac{\dot{N}}{\Delta X^2} \cdot \frac{\Delta X}{V} = \frac{\dot{N}}{V\,\Delta X} \qquad [\text{impacts/mm}^2\text{/pass}]. \tag{6}$$

This is exactly the formula of cell B13: `=B3/B8*(B11/B12)/(B11^2)`.

Numerical value: $R = \dfrac{6.43 \times 10^6}{300 \times 20} \approx 1072$ impacts/mm^2/pass.

Physical interpretation: $V\,\Delta X$ is the area swept per second (mm^2/s). R is simply "number of shots per second" divided by "area swept per second". The square shape of the jet therefore does not matter: only the *width* of the swept band counts.

3.4 Fraction of surface impacted per pass: the coefficient K

Each impact leaves a circular dent of radius r and area πr^2. The total area of the dents deposited during one pass, per unit surface, is:

$$K = \pi r^2\, R. \tag{7}$$

Numerical value: $K = \pi\,(0.010)^2 \times 1072 \approx 0.337$ (cell B17; the radius is converted from µm to mm by the factor 10^{-3}).

K is *not* the coverage itself: dents land at random and overlap, so the surface actually covered is smaller than K. This is the subject of the next paragraph.

3.5 Random coverage law (Poisson / Avrami)

Impacts are assumed to be randomly and independently distributed over the surface. After n passes, the number of dents covering a given point follows a Poisson law with mean $\lambda = K\,n$. The probability that a point has *never* been hit is the probability of zero events:

$$P(\text{not hit}) = e^{-\lambda} = e^{-Kn}. \tag{8}$$

The covered fraction is the complement, which gives equation (1):

$$\tau(n) = 100\left(1 - e^{-Kn}\right).$$

This is the formula of column G: `=100*(1-EXP(-$B$17*F3))`.

Remarks:

- For one pass and a small K, $\tau \approx 100\,K$: overlaps are negligible.
- Each additional pass only covers a fraction $(1 - e^{-K})$ of the surface *still uncovered*: the curve is concave and tends to 100 % without ever reaching it.

4 Results for the spreadsheet parameters

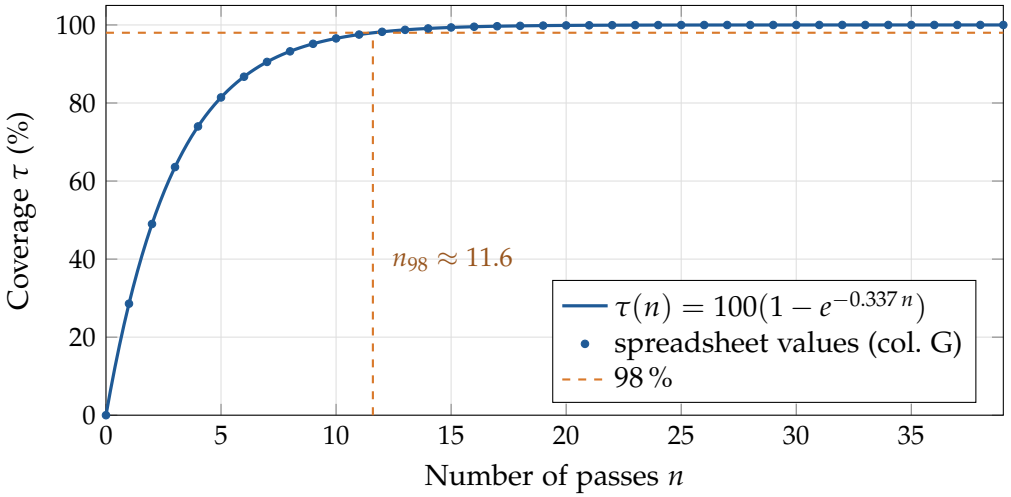


Figure 2: Predicted coverage versus number of passes ($K = 0.337$).

Table 2: Computed values (extract of column G).

n	1	2	3	4	6	8	10	12
τ (%)	28.6	49.0	63.6	74.0	86.7	93.2	96.6	98.2

4.1 Number of passes required for a target coverage

Inverting equation (1), the number of passes required for a target coverage τ^* is:

$$n^* = \frac{-\ln\left(1 - \tau^*/100\right)}{K}. \tag{9}$$

- $\tau^* = 98\,\%$ (usual threshold for "full coverage" in shot peening): $n^* = \ln 50/K = 3.91/K \approx 11.6$ passes.
- $\tau^* = 90\,\%$: $n^* = 2.30/K \approx 6.8$ passes.
- $\tau^* = 50\,\%$: $n^* = 0.69/K \approx 2.1$ passes.

A practical consequence: reaching 98 % requires about *1.7 times* more passes than 90 %, and going from 98 % to 99.9 % requires another 1.8 times more.

5 Influence of the parameters

The compact form (3) shows directly how each parameter acts on K, and therefore on how fast coverage builds up:

$$K = \frac{6\,\dot{m}\,r^2}{\rho\,D^3\,V\,\Delta X}.$$

Table 3: Sensitivity of K (and of $n_{98} \propto 1/K$) to each parameter, the others being fixed.

Parameter	Dependence of K	Effect when the parameter is doubled
Flow rate $\dot{m}$	$\propto \dot{m}$	$K \times 2$: half as many passes.
Dent radius r	$\propto r^2$	$K \times 4$: a quarter as many passes.
Density ρ	$\propto 1/\rho$	$K/2$: fewer shots for the same mass flow rate.
Diameter D	$\propto 1/D^3$	$K/8$ *at constant dent radius* (see § 6).
Traverse speed V	$\propto 1/V$	$K/2$: twice as many passes.
Jet size ΔX	$\propto 1/\Delta X$	$K/2$: the same flow is spread over a band twice as wide.

V and ΔX only appear through their product $V\,\Delta X$ (area swept per second): at constant media flow rate, what matters is the number of shots sent per unit of swept area.

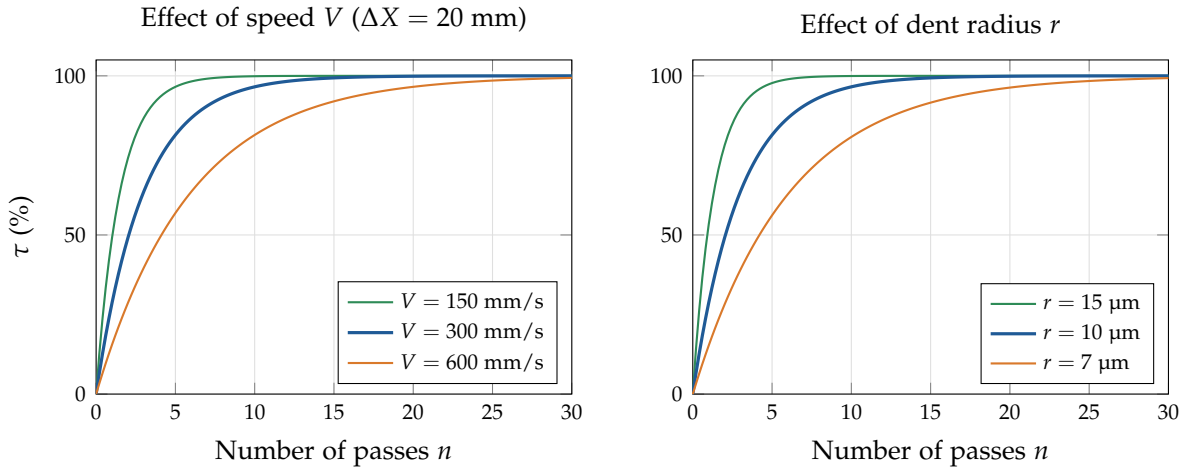


Figure 3: Sensitivity of the coverage curve to traverse speed and dent radius (other parameters: values of table 1).

6 Assumptions and limitations

1. **Uniform jet.** Shots are uniformly distributed within the jet. A real jet rather has a Gaussian-like profile (denser in the centre); coverage is then higher in the centre of the band than at its edges.

2. **One pass = one passage of the jet over each point.** The model does not handle the lateral step between two sweep lines: lines are implicitly assumed to be juxtaposed (step $= \Delta X$), with neither overlap nor gap. If lines overlap with a step $p < \Delta X$, replace ΔX by p in (3).

3. **Every shot hits and marks.** No media loss, rebound or broken shots (the nominal flow is fully effective). An efficiency $\eta < 1$ can be accounted for by multiplying K by η.

4. **Identical, independent dents.** All dents have the same radius r and impacts are random and independent (Poisson assumption). The effect of the media size distribution is not represented here (see the simulator technical note).

5. **Dent radius independent of the media.** r is a fixed input. In practice, increasing D also increases r (at equal impact velocity, r grows roughly in proportion to D). The $1/D^3$ sensitivity therefore only holds at fixed r; if $r \propto D$, then $K \propto 1/D$.

6. **Flat surface**, jet normal to the surface (no incidence-angle effect).

7 Correspondence with the spreadsheet

Cell	Excel formula	Meaning
B3	`=B2/60`	Flow rate in kg/s
B5	`=B4/1000*10^6`	Density in kg/m^3
B7	`=B6/10^6`	Diameter in m
B8	`=B5*4/3*PI()*(B7/2)^3`	Mass of one shot m_p (kg)
B10	`=B3/B8`	Shot flow rate $\dot{N}$ (shots/s)
B13	`=B3/B8*(B11/B12)/(B11^2)`	Impact density R (impacts/mm^2/pass)
B17	`=PI()*(B16*10^-3)^2*B13`	Coefficient K (r converted to mm)
G3:G42	`=100*(1-EXP(-$B$17*F3))`	Coverage $\tau(n)$ in %

Minor note: the label of cell A10 reads "nombre media/min" (shots per minute) while the computed value (B10) is in shots per *second*, as correctly stated by the unit in C10. The calculation is consistent; only the label needs correcting.

Dr Régis Kubler, Arts et Métiers Institute of Technology. Developed with the assistance of Claude Opus 5.5 (Anthropic), October 2026. Educational model with simplifying assumptions; it does not replace coverage verification according to the applicable standards (e.g. SAE J2277). Licence: CC BY 4.0.